Inorganic CUME
April-29-2016

Answer any 5 questions (but only 5)

1. (20 points) Tantalum (Ta) is packed to form a body-centered cubic (bcc, see Figure 1). The edge length of the unit cell of Ta, is 3.31Å. Tantalum has a density of 16.69 g/cm^3. Calculate the number of atoms in each unit cell and the mass of a tantalum atom. (Note: 1 Å= 10^{-8} cm).

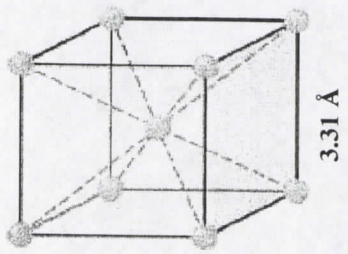

Figure 1. BCC structure of Tantalum

2. (20 points) Calculate the d-spacing corresponding to the (110) plane of Ta. (Note: a=b=c= 3.31 Å for Ta).

3. (20 points) Is it possible to solve the crystal structure of Ta completely using powder X-ray alone or is single crystal diffraction data required? Explain your answer.

4. (20 points) Based on the chemistry of the surrounding elements in what oxidation state or form would you expect tantalum to be most commonly found?

5. (20 points) Prussian blue can be made by the addition of Fe(II) to $K_3Fe(CN)_6$ or Fe(III) to $K_4Fe(CN)_6$. All of these reagents are weakly colored. Prussian blue however is an intense blue color and one of the earliest synthetic pigments. Describe the reaction responsible for the formation of the solid called Prussian Blue, describe the structure of the solid and describe the electronic transition responsible for the blue color.

6. (20 points) $[Co(N_4)L_2]^{+3}$ where N_4 is a strong field neutral macrocyclic ligand and L is acetonitrile gives a well-defined 1H NMR spectrum. Addition of a one electron reductant results in a spectrum which covers a range well outside of the traditional 0-10 ppm window and the individual lines are broader and not easily assigned. Rationalize this observation. Be very specific.

7. (20 points) Lithium is a very active metal. Does Li have a body-centered or face-centered cubic structure? Lithium and its compounds have several industrial applications. Give an example for those applications.

8. (20 points) Zn metal and many other first row metals readily dissolve in acid and give hydrogen gas. Copper, however, is essentially nonreactive. Rationalize this observation.

9. (20 points) Iron pentacarbonyl, like many metal carbonyl compounds is remarkably photochemically active. Traditionally, $Fe(CO)_5$ is irradiated in glacial acetic acid to provide a product in nearly quantitative yield. Suggest a product and indicate how you would use IR spectroscopy to verify the product.

10. (20 points) $CpMn(CO)_3$ is among the most stable organometallic compounds known. It does not react even if refluxed in pure triphenylphosphine for several hours. Describe two reactions that take advantage of the thermal stability but lead to potentially useful derivatives. The reactions should be sufficiently general as to provide synthetic routes to other derivatives. Provide some text in support of your choice.